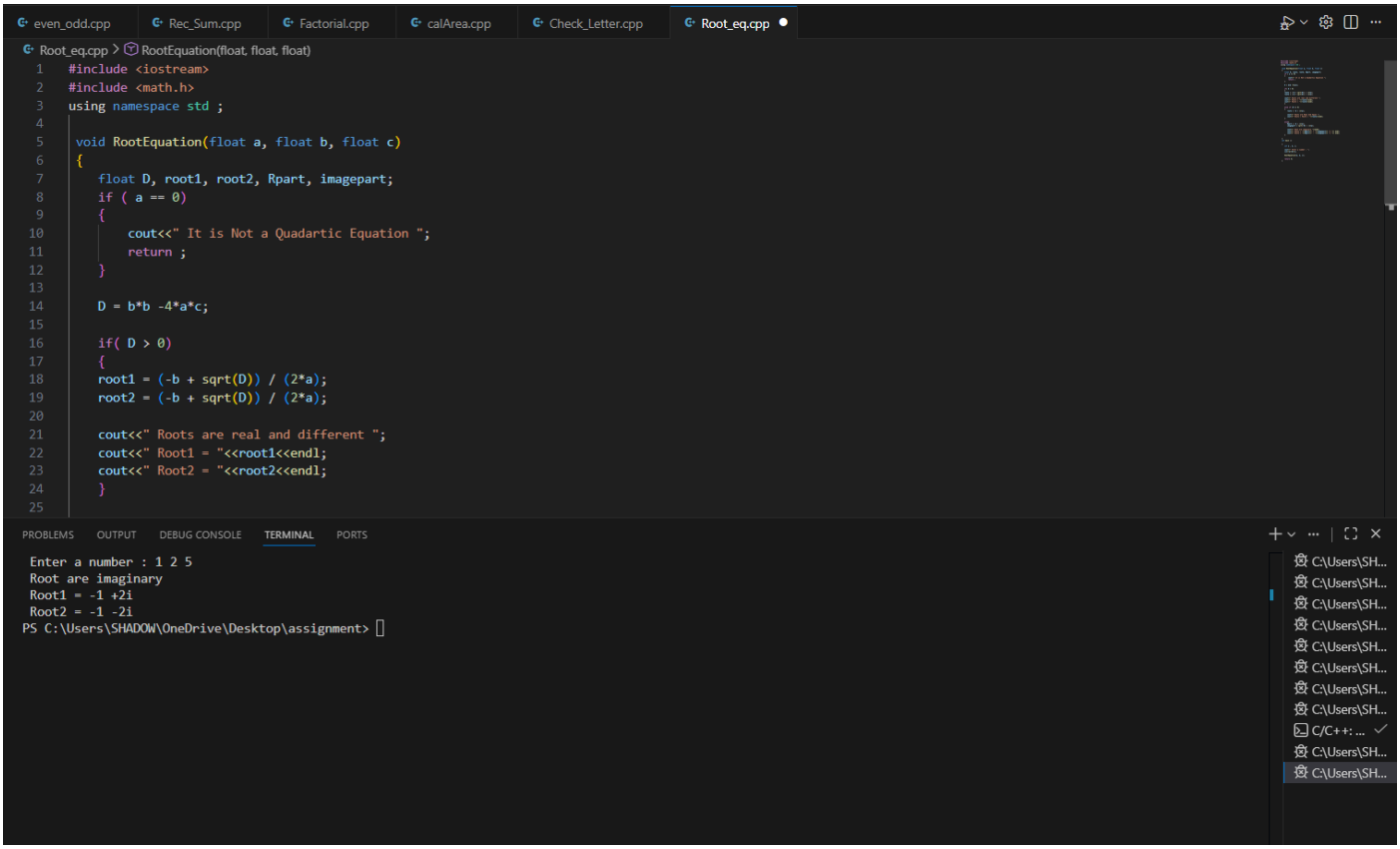
Q1] find Root of the Equation

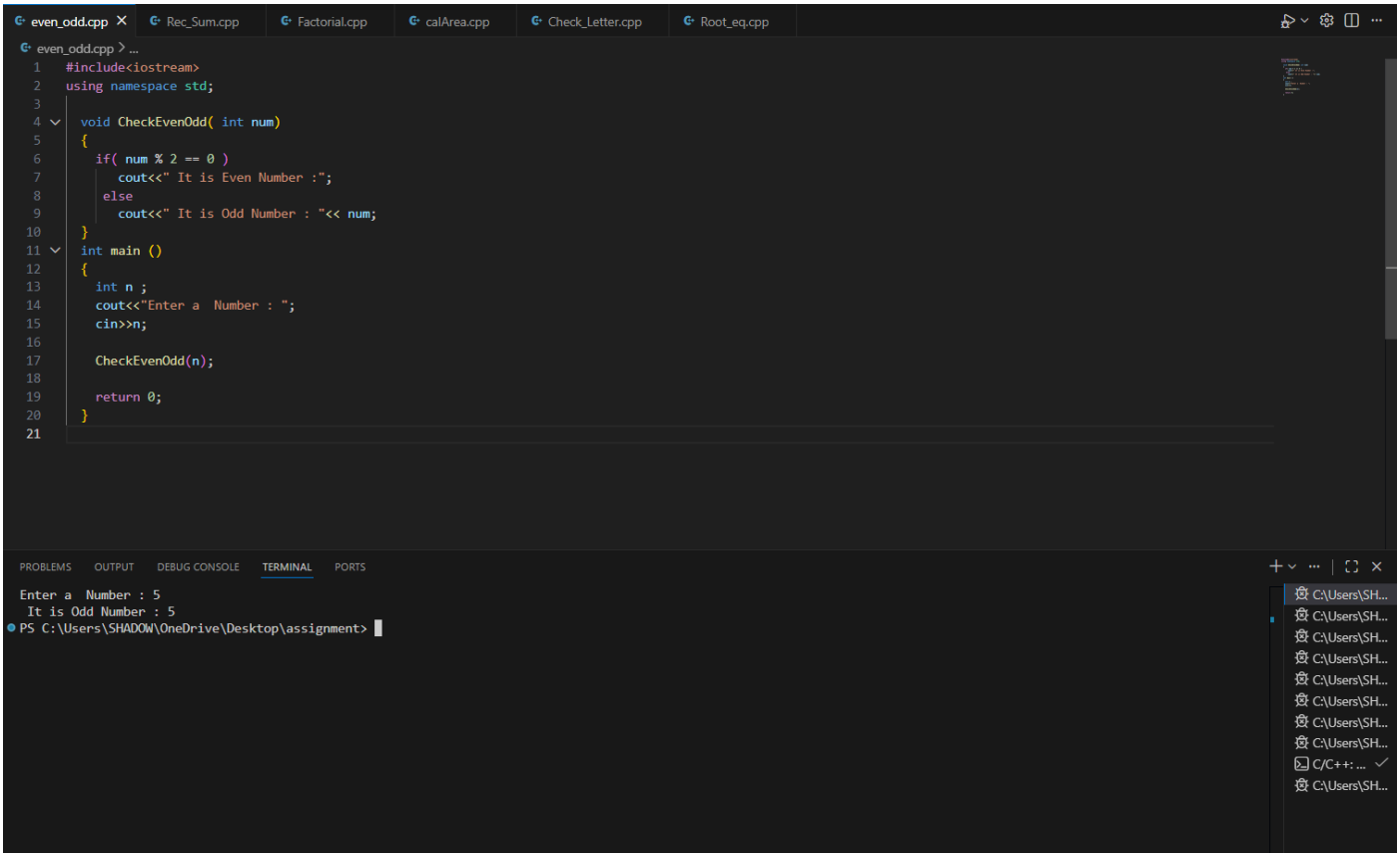
```
#include <iostream>
#include <math.h>
using namespace std ;

 void RootEquation(float a, float b, float c)
{
   float D, root1, root2, Rpart, imagepart;
   if ( a == 0)
   {
      cout<<" It is Not a Quadartic Equation ";
      return ;
   }

   D = b*b -4*a*c;

   if( D > 0)
   {
   root1 = (-b + sqrt(D)) / (2*a);
   root2 = (-b + sqrt(D)) / (2*a);

   cout<<" Roots are real and different ";
   cout<<" Root1 = "<<root1<<endl;
   cout<<" Root2 = "<<root2<<endl;
   }
   else if (D == 0)
   {
     root1 = -b / (2*a);
     cout<<" Roots are Real And Equal ";
     cout<<" Root1 = Root2 = "<<root1<<endl;
   }

   else{
     Rpart = -b / (2*a);
```

```
    imagepart = sqrt(-D) / (2*a);


    cout<<" Root are imaginary "<<endl;
    cout<<" Root1 = "<<Rpart<<" + "<<imagepart<< "i "<< endl;
    cout<<" Root2 = "<<Rpart<<" - "<<imagepart<< "i "<< endl;
  }
}
int main ()
{
  int a , b, c;
  cout<<" Enter a number : ";
  cin>>a>>b>>c;
  RootEquation(a, b, c);
  return 0;
}
```



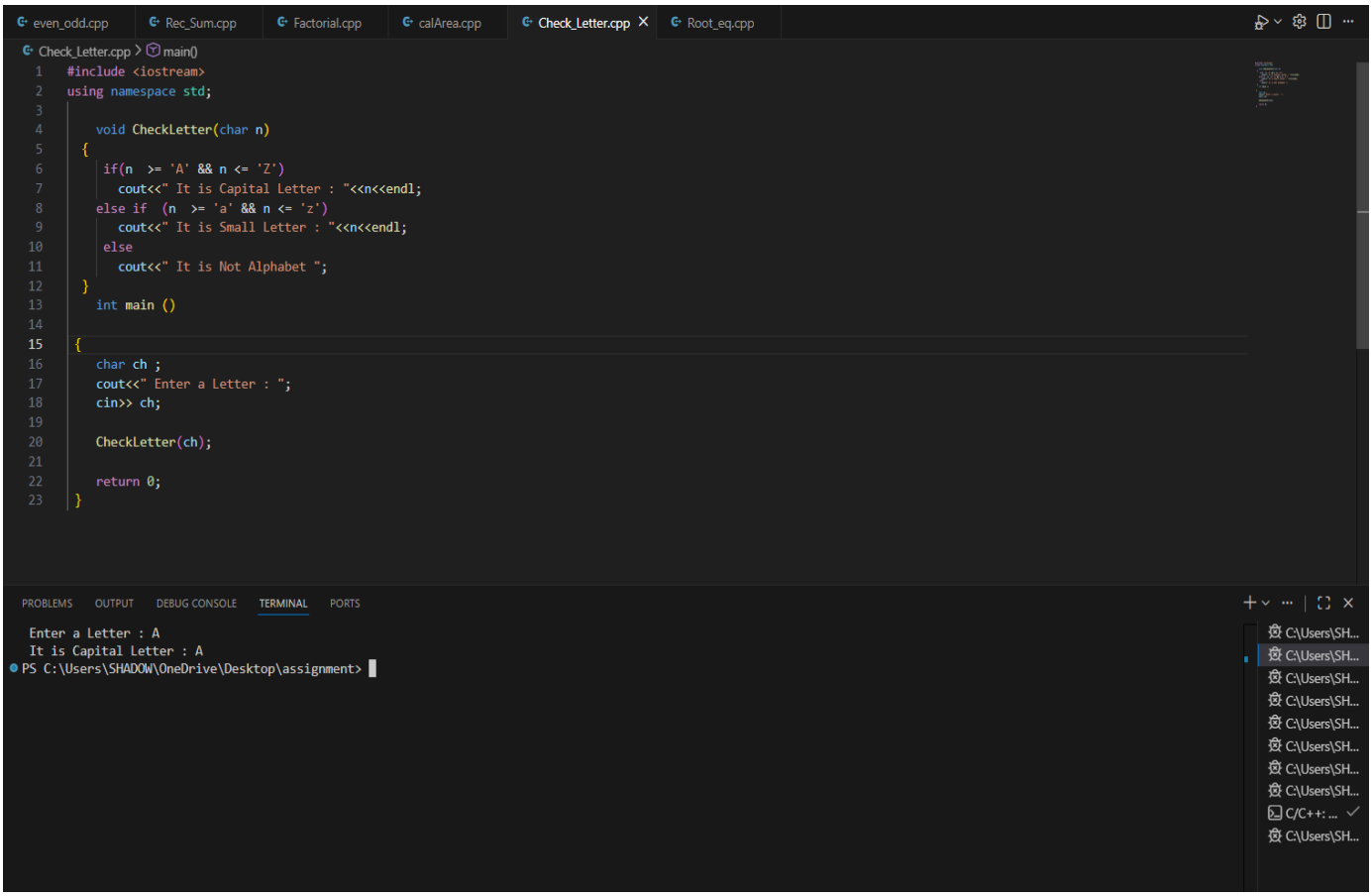
2. Check given number is odd or even number.

```
#include<iostream>
using namespace std;

 void CheckEvenOdd( int num)
 {
  if( num % 2 == 0 )
    cout<<" It is Even Number :";
   else
    cout<<" It is Odd Number : "<< num;
 }
 int main ()
 {
  int n ;
  cout<<"Enter a  Number : ";
  cin>>n;

  CheckEvenOdd(n);

  return 0;
 }
```



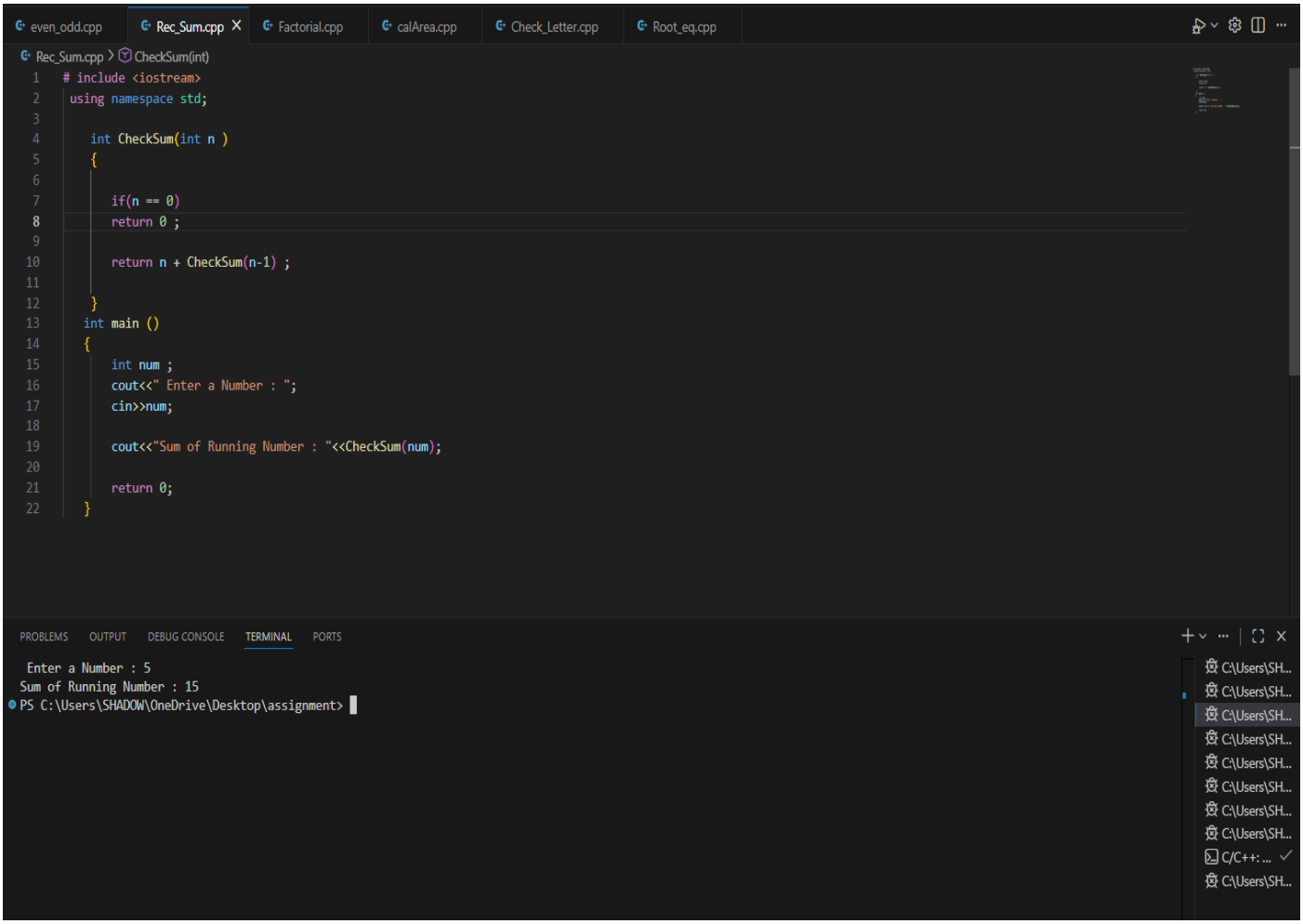
3. Check if the entered char is small letter or capital letter.

```
#include <iostream>
using namespace std;
  void CheckLetter(char n)
 {
   if(n  >= 'A' && n <= 'Z')
     cout<<" It is Capital Letter : "<<n<<endl;
  else if  (n  >= 'a' && n <= 'z')
     cout<<" It is Small Letter : "<<n<<endl;
   else
     cout<<" It is Not Alphabet ";
 }
  int main ()
{
  char ch ;
  cout<<" Enter a Letter : ";
  cin>> ch;
  CheckLetter(ch);
  return 0;
}
```



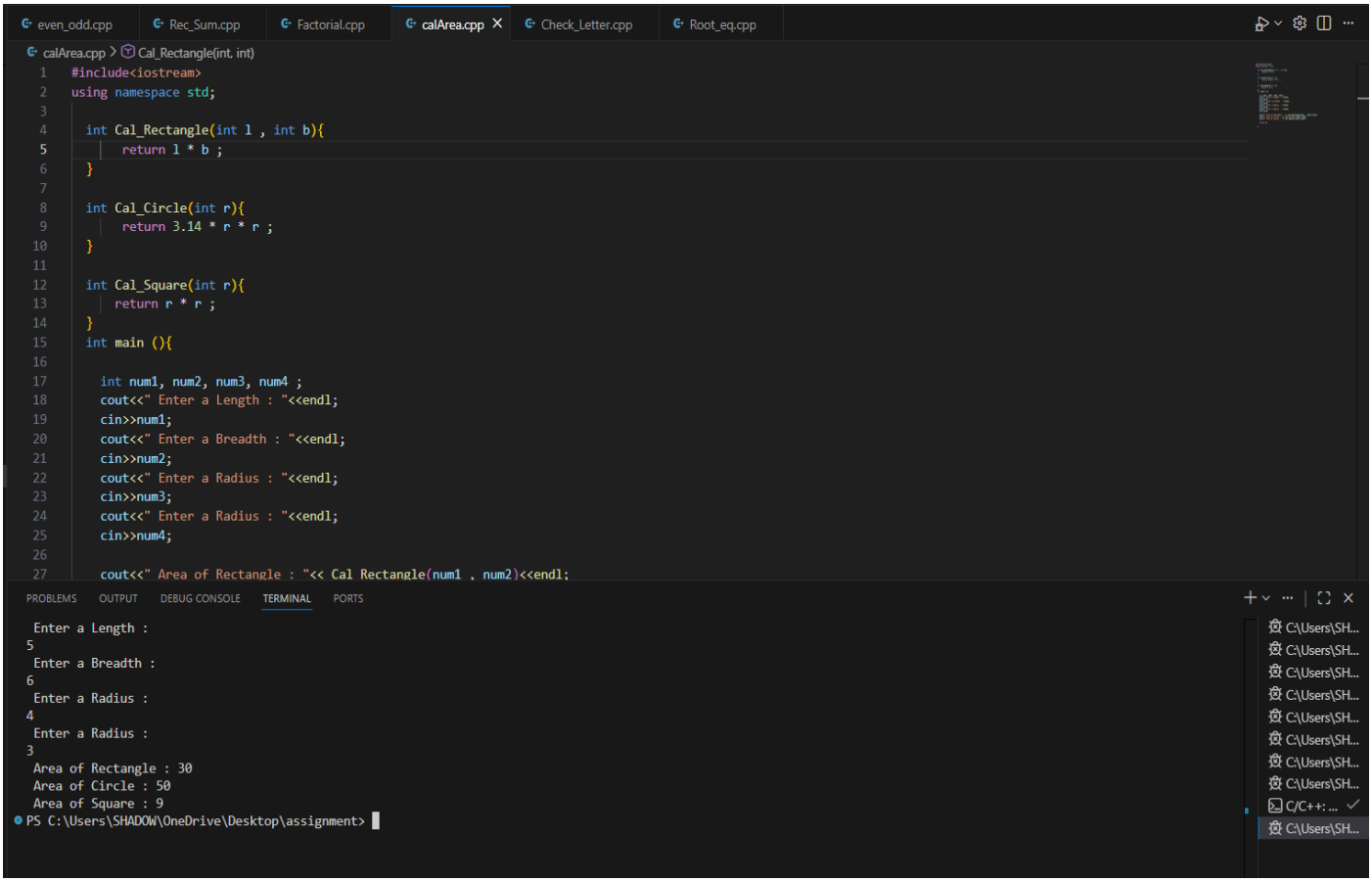
4. Write a recursive function to obtain the running sum of n numbers.

```
# include <iostream>
 using namespace std;
   int CheckSum(int n )
   {
     if(n == 0)
     return 0 ;
     return n + CheckSum(n-1) ;
   }
  int main ()
  {
     int num ;
     cout<<" Enter a Number : ";
     cin>>num;
     cout<<"Sum of Running Number : "<<CheckSum(num);
     return 0;
  }
```



5. Write functions for calculating area of circle, rectangle, and square and call these functions from main function.

```
#include<iostream>
using namespace std;
 int Cal_Rectangle(int l , int b){
     return l * b ;
 }
 int Cal_Circle(int r){
     return 3.14 * r * r ;
 }
 int Cal_Square(int r){
    return r * r ;
 }
 int main (){
  int num1, num2, num3, num4 ;
  cout<<" Enter a Length : "<<endl;
  cin>>num1;
  cout<<" Enter a Breadth : "<<endl;
  cin>>num2;
  cout<<" Enter a Radius : "<<endl;
  cin>>num3;
  cout<<" Enter a Radius : "<<endl;
  cin>>num4;
  cout<<" Area of Rectangle : "<< Cal_Rectangle(num1 , num2)<<endl;
  cout<<" Area of Circle : "<< Cal_Circle( num3)<<endl;
  cout<<" Area of Square : "<< Cal_Square(num4)<<endl;
  return 0;
 }
```



6. Write a function for factorial and use the function for printing factorial series like 1 2 6 24 120.....

```
#include<iostream>
using namespace std;
  int Fact(int n ){
    if (n == 0)
    return 1;
    return n * Fact(n-1);
  }
  int main ()
{
  int num ;
  cout<<" Enter a Number : ";
  cin>>num;
  cout<<" Factorial Number is : "<< Fact(num);

  return 0;
}
```

even_odd.cpp | Rec_Sum.cpp | Factorial.cpp | calArea.cpp | Check_Letter.cpp | Root_eq.cpp

Factorial.cpp > main()

```cpp
#include<iostream>
using namespace std;

   int Fact(int n ){
     if (n == 0)
     return 1;

     return n * Fact(n-1);
   }
   int main ()
 {
   int num ;
   cout<<" Enter a Number : ";
   cin>>num;

   cout<<" Factorial Number is : "<< Fact(num);

   return 0;

   }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
 Enter a Number : 5
 Factorial Number is : 120
PS C:\Users\SHADOW\OneDrive\Desktop\assignment>
```